Peter Carragher | Curriculum Vitae

Software and Societal Systems – Carnegie Mellon University – Pittsburgh, US

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Research Interests

Computational social science; network science; simulation modeling; media ecosystems; information diffusion; multimodal AI.

Education

Carnegie Mellon University **Pittsburgh, PA**
PhD in Societal Computing, GPA: 3.7, Advisors: Dr. Kathleen M. Carley, Dr. Patrick Park. *2022–present*
Research in computational social science, misinformation detection, webgraph analysis, and multimodal AI.

Georgia Institute of Technology **Atlanta, GA**
MSc in Computer Science, GPA: 3.7 *2021–2022*
Specialization in machine learning, reinforcement learning, knowledge-based AI, AI ethics, and network science. Contributed to a virtual teaching assistant for large-scale online learning with the Design & Intelligence Lab.

University of Edinburgh **Edinburgh, UK**
BSc in Computer Science, GPA: 4.0 *2014–2018*
Thesis: mapping DNN/CNN primitives to FPGA fabric via extended compiler support for heterogeneous architectures.

Industry Experience

Meta (Facebook) **London, UK**
Machine Learning Engineer *2020–2021*
Built detection systems that significantly reduced impersonation and identity fraud on Facebook and Instagram.

Software Engineer *2018–2019*
Built response systems to prevent coordinated inauthentic behavior targeting the 2019 Indian and EU elections. Deployed ML models to automate abuse reporting, saving ∼$10M/year.

Software Engineering Intern *2017–2018*
Added an SVG renderer to the augmented reality graphics engine; automated publication of scripting API documentation.

Codeplay **Edinburgh, UK**
Software Engineer, Part-time *2015–2016*
Worked on the LPGPU2 project funded by the *Horizon 2020* research and innovation programme. Built a cross-platform GPU profiling tool, which was subsequently presented at IWOCL and featured by the Khronos Group.

Fellowships and Awards

2024 – 2026: Knight Fellow: Transdisciplinary research fellowship

Publications

Journal Articles

Carragher, **Peter**, Evan M. Williams, and Kathleen M. Carley (Jan. 2025). "Misinformation Resilient Search Rankings with Webgraph-Based Interventions". In: *ACM Trans. Intell. Syst. Technol.* 16.1, 10:1–10:27.

Working Journal Papers

Marigliano, Rebecca, **Peter Carragher**, and Kathleen M. Carley (2026a). (2026a). *Modeling Emotional Expression with LLM Agents in a Synthetic Social Simulation*. Unpublished manuscript.

Marigliano, Rebecca, **Peter Carragher**, and Kathleen M. Carley (2026b). (2026b). *Simulating Opinion Dynamics with LLM-Generated Populations in Synthetic Social Networks*. Unpublished manuscript.

Teleki, Maria, Xiangjue Dong, **Peter Carragher**, Vedangi Bengali, Sai Tejas Janjur, Haoran Liu, Tian Liu, Cong Wang, Ting Liu, Yin Zhang, Frank Shipman, et al. (2026). "A Survey on LLMs for Story Generation". In.

Conference Proceedings

Carragher, **Peter** and Kathleen M. Carley (2026a). "Interaction Diversity in News Ecosystems: Accounting for Organizational Structure and Ownership". In: *Social, Cultural, and Behavioral Modeling*. Cham: Springer Nature Switzerland.

Williams, Evan M., **Peter Carragher**, Kyle Herdrich, Luke Prakarsa, and Kathleen M. Carley (2026). "Extending the BEND Framework to Webgraphs". en. In: *Social, Cultural, and Behavioral Modeling*. Cham: Springer Nature Switzerland, pp. 76–85.

King, Catherine, **Peter Carragher**, and Kathleen M. Carley (June 2025). "Mapping the Scientific Literature on Misinformation Interventions: A Bibliometric Review". en-US. In: *Workshop Proceedings of the 19th International AAAI Conference on Web and Social Media* 2025, p. 10.

Williams, Evan M., **Peter Carragher**, and Kathleen M. Carley (June 2025). "Bridging Social Media and Search Engines: Dredge Words and the Detection of Unreliable Domains". en. In: *Proceedings of the International AAAI Conference on Web and Social Media* 19, pp. 2030–2043.

Carragher, **Peter** and Kathleen M. Carley (2024). "Accountability in Search Engine Manipulation: A Case Study of the Iranian News Ecosystem". en. In: *Social, Cultural, and Behavioral Modeling*. Cham: Springer Nature Switzerland, pp. 154–163.

Carragher, **Peter**, Evan M. Williams, and Kathleen M. Carley (May 2024). "Detection and Discovery of Misinformation Sources Using Attributed Webgraphs". en. In: *Proceedings of the International AAAI Conference on Web and Social Media* 18, pp. 214–226.

Carragher, **Peter**, Lynnette Hui Xian Ng, and Kathleen M. Carley (2023). "Simulation of Stance Perturbations". en. In: *Social, Cultural, and Behavioral Modeling*. Lecture Notes in Computer Science. Cham: Springer Nature Switzerland, pp. 159–168.

Conference Papers Under Review

Carragher, **Peter** and Kathleen M. Carley (2026b). (2026b). *Measuring Organizational Influence on Digital News: An Information-Theoretic Framework for Media Pluralism and Editorial Independence*. Under review at The Web Conference (WWW) 2027. arXiv: 2610.40172.

– (2026c). (2026c). *Who Owns Digital News, and Who Gets Cited: Measuring Media Concentration Across 77 Countries and Seven AI Systems*. Under review at The Web Conference (WWW) 2027. Dataset: https://doi.org/10.5281/zenodo.22782826. arXiv: 2610.40164.

Workshop Papers

Birkelbach, Stephanie, Maria Teleki, **Peter Carragher**, Xiangjue Dong, Nehul Bhatnagar, and James Caverlee (May 2026). "SocialPulse: An Open-Source Subreddit Sensemaking Toolkit". In: *Workshop Proceedings of the 20th International AAAI Conference on Web and Social Media*. arXiv:2602.07248 [cs].

Irmetova, Anna, Haoran Liu, Maria Teleki, **Peter Carragher**, Julie Zhang, and James Caverlee (May 2026). "PodChecker: An Interpretable Fact-Checking Companion for Podcasts". en. In: *Workshop Proceedings of the 20th International AAAI Conference on Web and Social Media*.

Carragher, **Peter**, Abhinand Jha, Raghav R, and Kathleen M. Carley (Aug. 2025). "Quantifying Memorization and Parametric Response Rates in Retrieval-Augmented Vision-Language Models". In: *Proceedings of the First Workshop on Large Language Model Memorization (L2M2)*. Vienna, Austria: Association for Computational Linguistics, pp. 127–141.

Carragher, **Peter**, Nikitha Rao, Abhinand Jha, R. Raghav, and Kathleen M. Carley (June 2025). "SegSub: Evaluating Robustness to Knowledge Conflicts and Hallucinations in Vision-Language Models". en-US. In: *Workshop Proceedings of the 19th International AAAI Conference on Web and Social Media* 2025, p. 27.

Software

Demos

2026: *NetNeighbors: Webgraph Explorer*. https://netneighbor.petercarragher.com/

2025: *NewsGuess: News Site Ownership Guessing Game*. https://newsguess.petercarragher.com/

2025: *NewsMap: Global News Networks Map*. https://newsmap.petercarragher.com/

Open Source Contributions

2026: *pyccwebgraph: Python Interface to CommonCrawl Webgraph*. PyPI: https://pypi.org/project/pyccwebgraph/

2026: *cc-pyspark: Process Common Crawl Data with Python and Spark*. GitHub: https://github.com/commoncrawl/cc-pyspark. Contributor

Talks

2025: *A Hitchhiker's Guide to Adversarial Adaptation In Information Systems*. AI and Disinformation Summit, IDeaS Center for Social Cybersecurity, Carnegie Mellon University, Pittsburgh, PA

2025: *Deploying Large Scale Trust & Safety Systems*. Managing Disruptive Technologies (95-723), Heinz College, Carnegie Mellon University

2025: *Measuring the Effect of Ownership Concentration on News Publication Practices*. IDeaS Summit 2025, Carnegie Mellon University, Pittsburgh, PA. October 15–17, 2025

2025: *Measuring the Effect of Ownership Concentration on News Publication Practices*. IC2S2 2025: 11th International Conference on Computational Social Science, Norrköping, Sweden. Extended abstract, non-archival

2024: *Disinformation Research Presentation and Tool Demo*. Disinformation & Democracy Workshop, IDeaS Center for Social Cybersecurity, Carnegie Mellon University, Pittsburgh, PA

Teaching and Mentorship

2026: *Teaching Assistant (Fall 2026)*. Computational Modeling of Complex Socio-Technical Systems (17-821/17-621), Carnegie Mellon University. Instructor: Prof. Kathleen M. Carley

2025: *Undergraduate Research Mentor*. REUSE Program, Carnegie Mellon University

2025: *Computer Science Pedagogy Student*. 15-890, Carnegie Mellon University. Instructors: Michael Hilton, Franceska Xhakaj. Developed curriculum for future courses on misinformation and trust and safety systems; prepared teaching philosophy and portfolio

2025: *Instructor*. OMEN Exercise (Operational Mastery of the Information Environment). Led training sessions on misinformation detection and countering

2024: *Volunteer Organizer*. S3D Lunch & Learn Series, Carnegie Mellon University (2 years). https://s3d.cmu.edu/events/lunch-and-learn.html

2023: *Instructor*. CASOS Summer Institute, Carnegie Mellon University. Delivered modules on network robustness measures, stance detection, misinformation detection, and LLM hallucinations. Summer '23, '24, '25, & '26.

2022: *Teaching Assistant*. Knowledge-Based AI (CS7637), Georgia Institute of Technology. Instructor: Prof. Ashok Goel. Spring 2022

2021: *Student of CS 6460, Educational Technology*. Georgia Institute of Technology. Instructor: Dr. David Joyner. Research project deploying Jill Watson virtual TA on Piazza for large-scale online learning

Professional Service

2026: *Workshop Organizer*. 2nd Workshop on Misinformation Detection in the Era of LLMs (MisD), co-located with ICWSM 2026, Los Angeles, CA

2026: *Panel Host*. Conference Panel on AI Governance, SBP-BRiMS 2026

2026: *Reviewer*. COLM Workshop: Large Language Model Memorization (L2M2 2026)
2026: *Reviewer (Program Committee)*. International AAAI Conference on Web and Social Media (ICWSM 2026)
2026: *Reviewer (Program Committee)*. International Conference on Computational Social Science (IC2S2 2026)
2025: *Admissions Committee*. REUSE Program Admissions, Carnegie Mellon University
2025: *Reviewer*. ACL Workshop: Large Language Model Memorization (L2M2 2025)
2025: *Seminar Organizer*. Computational Social Science Seminar, Carnegie Mellon University
2025: *Student Volunteer*. ICWSM 2025, Copenhagen, Denmark
2024: *Reviewer*. Information Democracy & Social – cybersecurity Conference (IDeaS 2024)